

# Bounds for identifiability of sparse and dense Erdős–Rényi graphs

Claire Mao

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## Abstract

We discuss upper and lower bounds on radius  $r$  for the identifiability of an Erdős–Rényi random graph from its  $r$ -neighborhoods.

## 1 Introduction

The shotgun assembly problem has several applications, from DNA sequencing to understanding genetic diseases to reconstructing large neural networks [2]. The generalized problem is that we are given many small pieces of a whole (sub-sequences of DNA, subnetworks of a neural network, or subgraphs of a graph) and want to reconstruct the original. In graph theory, the shotgun assembly problem takes on a specialized form: we wish to use subgraphs of usually random graphs to identify and reconstruct the original.

One popular way to generate random graphs is the Erdős–Rényi model, first introduced in 1959. Let  $G(n, p)$  be an Erdős–Rényi graph with  $n$  vertices and edge probability  $p$ ; then, each edge appears in  $G$  with probability  $p$ . In graph shotgun assembly, we are interested in reconstructing such graphs from neighborhoods of radius  $r$ .

A neighborhood of radius  $r$ , denoted  $\mathcal{N}_r(v)$ , is the labeled subgraph induced by vertices within distance at most  $r$  of  $v$ . Thus a graph  $G$  with  $n$  vertices has  $n$   $r$ -neighborhoods, one for each vertex.

As an example, we have the graph  $G$ :

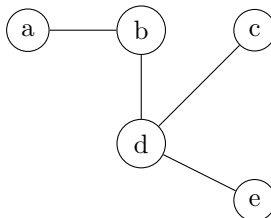


Figure 1: Graph  $G$

The following are the 1-neighborhoods of  $G$  around the vertices  $a$  and  $d$ , respectively.



Figure 2: 1-neighborhood of vertex  $a$

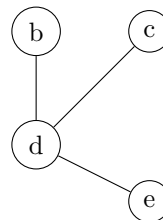


Figure 3: 1-neighborhood of vertex  $d$

Recall that graphs  $G$  and  $H$  are isomorphic if there is a bijection between the respective vertex sets  $V(G)$  and  $V(H)$  such that two vertices in  $G$  are adjacent in  $G$  if and only if they are adjacent in  $H$ .

There are two guiding questions for the shotgun assembly problem: identifiability and reconstruction. Identifiability is a prerequisite for reconstruction and will be the focus of this paper. We will follow the approach of Mossel and Ross [5] throughout this paper.

**Definition 1** (Identifiability). *Given each of the  $N$  neighborhoods  $N_r(v)$  for a vertex  $v$  in the graph, can we correctly identify  $G_N$  and its labels?*

**Definition 2** (Reconstruction). *For what range of  $p$  is a random graph  $G \in G(n, p)$  reconstructible from its  $r$ -neighborhoods with high probability?*

Note that we say that  $G$  is reconstructible from its  $r$ -neighborhoods if every graph with  $r$ -neighborhoods isomorphic to those of  $G$ , is itself isomorphic to  $G$  [4].

These two questions are still being answered as researchers produce tighter upper and lower bounds for  $r$  and work on applying our knowledge of the graph shotgun assembly problem to the ever-growing field of machine learning and neural networks.

## 1.1 Probability review

In order to take meaningful steps toward bounding  $r$ , we first familiarize ourselves with useful probability tools. We introduce the indicator and Bernoulli random variables, the notion of convergence in probability, and the Chernoff bound for bounding the tails of a random variable.

An indicator random variable takes value 1 if a certain event occurs, and 0 otherwise. A Bernoulli random variable takes value 1 with probability  $p$ , and takes value 0 with probability  $1 - p$ .

We denote that a sequence of random variables  $X_1, X_2, \dots$  converges in probability to a random variable  $X$ , denoted  $X_n \xrightarrow{p} X$ , if for every  $\varepsilon > 0$ ,  $\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0$ .

**Theorem 1** (Chernoff Bound). *Let  $X = \sum_{i=1}^n X_i$ , where each  $X_i = 1$  with probability  $p_i$  and  $X_i = 0$  with probability  $1 - p_i$ , and the  $X_i$  are independent. Let  $\mathbb{E}[X] = \mu$ . Then for  $0 < \epsilon < 1$ :*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \epsilon \mathbb{E}[X]) \leq 2 \exp\left(-\frac{\epsilon^2}{3} \mathbb{E}[X]\right).$$

## 1.2 Sparse and dense graphs

For this paper, we define a sparse Erdős-Rényi graph as one with  $N$  vertices and  $p_N = \lambda/N$  for fixed  $\lambda > 0$ . In other words, the average degree of a node in this graph remains bounded as  $N$  goes to infinity, and sparse graphs are sometimes referred to as bounded average degree graphs.

Sparse and dense graphs differ in their respective number of edges, with sparse graphs having fewer edges than their dense counterparts. In random graphs, the density of the graph is determined by the edge probability – the higher the edge probability, the more dense the graph is likely to be. If we return to the main idea of using neighborhoods to identify a graph, we understand how density affects identifiability.

## 2 Blocking configurations

A blocking configuration in a graph is a structure or pattern that prevents identifiability in that graph. This is because blocking configurations hinder uniqueness, causing two (or more) different vertices  $v$  and  $w$  in the graph to have isomorphic  $r$ -neighborhoods ( $\mathcal{N}_r(v) \cong \mathcal{N}_r(w)$ ). Thus, the  $r$ -neighborhoods of the graph will not contain enough unique information to reconstruct the original graph up to isomorphism. These configurations are tools that allow us to find the lower bound on neighborhood size for identifiability.

The following Lemma and Proposition, proposed by Mossel and Ross [5], guide us in our first steps toward lower bounding  $r$  using blocking configurations and the idea of a shell.

Given graph  $G$ , we define the shell  $S(v; s, t)$  as the subgraph formed by removing all isolated vertices from the subgraph of  $G$  induced by the vertices  $u$  with  $s \leq d(u, v) \leq t$  for  $0 < s < t$  and vertex  $v$  of  $G$ . Then:

**Lemma 1.** *Suppose  $G$  is a graph such that there exists  $r > 0$  and vertices  $v, w$  satisfying:*

1.  $S(v; 1, 2r) = S(w; 1, 2r)$ ,
2.  $d(v, w) > 2r$ ,
3. the graph obtained by switching  $\mathcal{N}_1(v)$  and  $\mathcal{N}_1(w)$  in  $G$  is not isomorphic to  $G$ ,

then identifiability from  $r$ -neighborhoods is impossible.

Proof of Lemma 1 is omitted from this paper. An example of a graph satisfying the three conditions of this lemma is provided below; we can verify this by setting  $v$  to be the leftmost vertex on the path graph and  $w$  to be one of the vertices of degree 3.

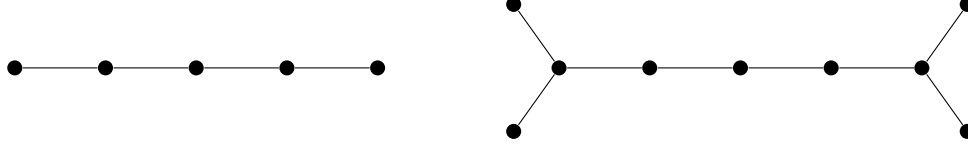


Figure 4: Example  $G$  satisfying Lemma 1

**Proposition 1.** Consider the Erdős–Rényi graph on  $N$  vertices with edge probability  $p_N = \lambda/N$ . Taking the limit as  $N \rightarrow \infty$ :

- If  $\sqrt{N}\lambda^r(1 - \lambda/N)^{Nr} \rightarrow \infty$ , then the probability of identifiability tends to zero.
- If  $\lim_{N \rightarrow \infty} \sqrt{N}\lambda^r(1 - \lambda/N)^{Nr} > 0$ , then the probability of identifiability is strictly less than one.

We sketch the proof of this proposition. We define a blocking configuration composed of two components: a path graph with  $2r + 1$  vertices, and a component with  $2r + 5$  total vertices: a  $2r + 1$ -length path graph and 2 additional vertices attached to both ends of the path, forming “prongs”.

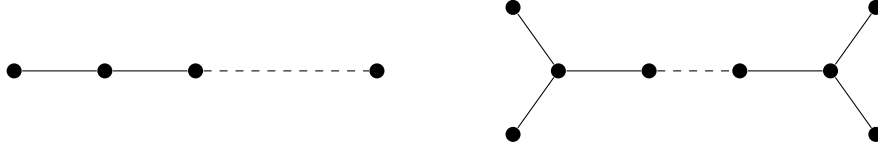


Figure 5: Blocking configuration defined for proof of Proposition 1

One can see that this blocking configuration satisfies the conditions of Lemma 1 by taking  $v$  as one of the endpoints of the path graph and  $w$  as one of the vertices with degree 3 in the pronged graph.

Then, we show that such a blocking configuration appears in the graph with high probability, and conclude that blocking configurations appear with high probability using various combinatorial techniques. From Proposition 1, we notice that  $r$  must be at least  $\sim \log(N)$  for identifiability to be possible. Of course, if  $r$  is too small, then blocking configurations appear with high probability and thus identifiability is less likely.

However, Proposition 1 does not apply to all Erdős–Rényi graphs, but only those with edge probability  $\lambda/N$ . Graphs with this edge probability are known as “sparse,” and this property of their edge density holds key implications for our desired  $r$ .

### 3 Sparse Erdős–Rényi graphs

Sparse graphs have a higher chance of being disconnected. Thus, neighborhoods of vertices in a sparse graph may contain less information than neighborhoods of vertices in a dense graph, and we require

a larger  $r$  on the order of  $\log(N)$  to identify a sparse graph with high probability. This result of  $r \sim \log(N)$  follows from Proposition 1.

As these two categories of random graphs are quite different, we find bounds for  $r$  for sparse and for dense graphs separately, and focus on sparse graphs in this section. First, we determined for what values of  $r$  identifiability is not possible (Lemma 1). Then, we utilized this result to estimate how large  $r$  should be to even have a chance of identifying the original graph from  $r$ -neighborhoods (Proposition 1).

We now define the diameter of a graph. If  $r$  is larger than the diameter of the graph, then clearly the graph is reconstructible from  $r$ -neighborhoods. Therefore, we claim the following properties about the diameter, and can relate them to properties of  $r$ .

**Theorem 2** (Diameter Theorem [5]). *Let  $G_N$  be the Erdős–Rényi random graph on  $N$  vertices with edge probability  $p_N = \lambda/N$  for a fixed  $\lambda > 0$ , and let  $D = D_{N,\lambda}$  be the maximum diameter of a component of  $G_N$ , i.e. the maximum distance between any two vertices in the component.*

- If  $\lambda < 1$ , then

$$\frac{D_{N,\lambda}}{\log(N)} \xrightarrow{p} \frac{1}{\log(1/\lambda)}.$$

- If  $\lambda = 1$ , then  $N^{-1/3}D_{N,1}$  converges in distribution to a non-negative and non-degenerate distribution (i.e. involving some randomness).
- If  $\lambda > 1$ , and  $\lambda_* < 1$  is the unique solution to  $\lambda e^{-\lambda} = \lambda_* e^{-\lambda_*}$ , then

$$\frac{D_{N,\lambda}}{\log(N)} \xrightarrow{p} \frac{1}{\log(\lambda)} + \frac{2}{\log(1/\lambda_*)}.$$

This factor of  $\log(N)$  appeared in the result of Proposition 1 as the lower bound for how  $r$  should grow. This is not a coincidence! We do not need  $r$  to be much larger than the typical maximum diameter of a graph component because we do not need the excess information provided by a neighborhood with radius beyond the typical maximum diameter. Thus, we treat the diameter and our results surrounding it as tools to help us upper bound  $r$ .

These different values of edge density parameter  $\lambda$  correspond to different regimes which describe what the graph looks like [3].  $\lambda < 1$ , the subcritical regime, is composed of many small graph components, sometimes described as trees.  $\lambda = 1$  is the critical regime, when a dominant graph component begins to form amid the smaller components. Finally,  $\lambda > 1$  is the supercritical regime, where one large unique graph component exists alongside many smaller tree components. We refer the reader to Chapters 1 and 2 of Durrett [1] for more detailed analysis of the Erdős–Rényi phase transition. Notice that both the subcritical and supercritical regime result in  $\frac{D_{N,\lambda}}{\log(N)}$  converging in probability to a constant term (because  $\lambda$  is fixed). In fact, we see that  $D_{N,\lambda}$  converges to a constant factor of  $\log(N)$  in both these regimes.

Following the definitions of regimes of  $\lambda$ , we reach our main theorem for sparse graphs which provide us with both an upper and lower bound, a direct result from the Proposition 1 and the Diameter Theorem.

**Theorem 3** (Identifiability Theorem [5]). *For the Erdős–Rényi graph on  $N$  vertices with  $p_N = \lambda/N$  for fixed  $\lambda > 0$  and taking limits as  $N \rightarrow \infty$ , if for some  $\varepsilon > 0$*

$$\frac{r}{\log(N)} < \frac{1}{2(\lambda - \log(\lambda))} - \varepsilon,$$

*then the probability of identifiability from  $r$ -neighborhoods tends to zero.*

- If  $\lambda < 1$  and for some  $\varepsilon > 0$ ,

$$\frac{r}{\log(N)} > \frac{1}{\log(1/\lambda)} + \varepsilon,$$

*then the probability of identifiability from  $r$ -neighborhoods tends to one.*

- If  $\lambda > 1$  and  $\lambda_* < 1$  is the unique solution to  $\lambda e^{-\lambda} = \lambda_* e^{-\lambda_*}$ , and for some  $\varepsilon > 0$ ,

$$\frac{r}{\log(N)} > \frac{1}{\log(\lambda)} + \frac{2}{\log(1/\lambda_*)} + \varepsilon,$$

then the probability of identifiability from  $r$ -neighborhoods tends to one.

Note that we have achieved the lower bound from Proposition 1, which provides conditions for which identifiability is impossible (i.e.  $r$  is too small), and the upper bound from the Diameter Theorem, as the maximum radius of a component of  $G$  converges to a constant multiple of  $\log(N)$  when  $\lambda \neq 1$ , and  $r$  is upper bounded by the typical diameter  $D$  of  $G$  as described above.

## 4 Dense Erdős–Rényi graphs

We define the dense Erdős–Rényi graph as a graph with  $N$  vertices and  $p_N \gg \log(N)^2/N$ . As  $N$  goes to infinity, the average degree goes to infinity, in contrast with the sparse definition.

To begin approaching a bound for this dense category of graphs, we make the following claim:

**Lemma 2.** *If  $X$  is a binomial distribution and  $x > 0$ , then*

$$\mathbb{P}(X > x) \leq \left(\frac{e}{x}\right)^x (\mathbb{E}X)^x.$$

Next, we claim that if all  $r-1$ -neighborhoods of a graph are unique, then the graph is identifiable from  $r$ -neighborhoods:

**Lemma 3.** *If  $\mathcal{N}_{r-1}(v) \neq \mathcal{N}_{r-1}(w)$  for all vertices  $v \neq w$ , then there is an algorithm for recovering the graph from  $r$ -neighborhoods.*

Though we will not prove this result here, one can reason that once  $r-1$ -neighborhoods are unique, then neighborhoods that are slightly larger (i.e. of size  $r$ ) are enough to identify the graph. Similar to blocking configurations, this tool of having unique neighborhoods of a certain size helps us determine a lower bound for  $r$  to ensure identifiability.

Moving back to the specific case of dense Erdős–Rényi graphs, we use Lemmas 2 and 3 to prove the 3-neighborhood Theorem, our main result for identifiability of dense graphs. This result is significant as it tells us that with high probability, dense Erdős–Rényi random graphs are identifiable from 3-neighborhoods.

**Theorem 4** (3-neighborhood Theorem [5]). *Let  $G$  be the Erdős–Rényi graph with  $N$  vertices and edge probability  $p_N$  such that as  $N \rightarrow \infty$ ,  $N * p_N / \log(N)^2 \rightarrow \infty$ . If we are given  $\mathcal{N}_3(v)$  for each vertex  $v$  in  $G$ , then the probability of identifiability tends to one.*

*Proof.* The following proof is adapted from Mossel and Ross [5].

We may assume, without loss of generality, that  $p_N \leq N^{-3/5}$ . This is because if  $p_N > N^{-3/5}$ , then with high probability the diameter of graph  $G$  is at most 3, and thus the result follows trivially from our discussion on how  $r$  need not be much greater than the diameter.

Let  $E$  be the event that each vertex  $v$  in  $G$  has a distinct 2-neighborhood  $\mathcal{N}_2(v)$ . In this proof, we first want to show that  $\mathbb{P}(E)$  tends to one. Then, we will apply Lemma 3 to show our desired result.

In order for event  $E$  to occur, no two vertices  $v$  and  $w$  may have  $\mathcal{N}_2(v) = \mathcal{N}_2(w)$ . This is our “bad event” which we want to bound by breaking this into two conditions which must hold. For  $\mathcal{N}_2(v) = \mathcal{N}_2(w)$ , we need:

- 1)  $\deg(v) = \deg(w)$ . (the degrees must match)
- 2)  $\{\deg(v_i) : v_i \in \mathcal{N}_1(v)\} = \{\deg(w_i) : w_i \in \mathcal{N}_1(w)\}$ . (the multisets of neighbors of  $v$  and  $w$  must match)

### 4.1 Bounding the probability that $\deg(v) = \deg(w)$

First, we bound the probability that  $\deg(v) = \deg(w)$ . Let  $\deg(v) = B_v + I$ , and let  $\deg(w) = B_w + I$ , where  $I$  is the indicator random variable that takes value 1 if  $v, w$  are neighbors. Without loss of generality,  $B_v \sim \text{Bin}(N-2, p_N)$  as there are  $N-2$  vertices excluding both  $v$  and  $w$ , and each of these vertices connect to  $v$  with probability  $p_N$ .

Applying the Chernoff Bound from Lemma 2 to this binomial distribution, we obtain that for all  $\epsilon_1 \in (0, 1/2)$ :

$$\mathbb{P}(\deg(v) \in Np_N(1 \pm \epsilon_1)) \geq 1 - 2 \exp \left\{ -\frac{\epsilon_1^2}{3} Np_N \right\}.$$

This gives us a lower bound for the probability of  $\deg(v)$  being “typical,” or around its expectation.

Next, we utilize this result and the fact that the event  $\deg(v) = \deg(w)$  is independent of indicator  $I$ . This means we can use the local limit theorem for binomial distributions, also known as the De Moivre-Laplace Theorem, to find an upper bound for the probability that  $\deg(v) = \deg(w)$  given that  $\deg(v)$  is near its expectation, or a typical degree. This gives us the following for some  $C$  that does not depend on  $N$ :

$$\mathbb{P}(\deg(w) = \deg(v) \mid \deg(v) \in Np_N(1 \pm \epsilon_1)) \leq \frac{C}{\sqrt{Np_N}}.$$

## 4.2 Bounding the probability $\deg(v_i)$ and $\deg(w_i)$ are equal as multisets

For two vertices  $v, w$  to have the same 2-neighborhood, their neighbors must have matching degree distributions, or matching multisets of degree counts (as multiplicity matters). To show that the probability of this event is small, we first account for overlaps between neighborhoods of  $v$  and  $w$ .

Recall that  $v_i$  are the neighbors of  $v$ , and  $w_i$  are the neighbors of  $w$ . We wish to analyze the overlap between the neighborhoods of  $v$  and  $w$ , given that  $\deg(v) = \deg(w)$ . Let us define the common degree when  $\deg(v) = \deg(w)$ :

$$M := \deg(v) \cdot \mathbf{I}[\deg(w) = \deg(v) \in Np_N(1 \pm \epsilon_1)],$$

and let

$$K := M - |\mathcal{N}_1(v) \cap \mathcal{N}_1(w)| - I$$

be the number of neighbors of  $v$  and  $w$  that are connected to exactly one of  $v$  or  $w$ .

We wish to understand the probability of neighbors of  $v$  and  $w$  being distinct rather than shared, as distinct neighborhoods will allow us to identify a graph with higher probability. We use a hypergeometric distribution, a discrete probability distribution which describes the probability of  $k$  successes in  $n$  trials without replacement.

For  $M > 0$ , we let the neighbors of  $v$  and  $w$  be chosen uniformly at random from  $N - 1$  potential neighbor vertices each. We conclude that  $M - K$  is hypergeometric, and because hypergeometric distributions can be represented as the sums of independent indicators, we apply the Chernoff bound. For further discussion, see Mossel and Ross [5]. Using the result, we have

$$\frac{1}{3}Np_N < K \leq M < 2Np_N.$$

Given  $M > 0$  and  $K$ , we define the multiset of degrees of the  $K$  neighbors that are connected to exactly one of  $v$  or  $w$ :

$$\{D(v)\} := \{D_1(v), \dots, D_K(v)\}, \quad \{D(w)\} := \{D_1(w), \dots, D_K(w)\}.$$

The results from above imply the following.

$$\mathbb{P}(\mathcal{N}_2(v) = \mathcal{N}_2(w)) \leq 2 \exp \left\{ -\frac{\epsilon_1^2}{3} Np_N \right\} + \frac{2C}{\sqrt{Np_N}} \exp \left\{ -\frac{Np_N}{6} (\epsilon_2 - 5p_N) \right\} + \frac{C}{\sqrt{Np_N}} \mathbb{P}(\{D(v)\} = \{D(w)\} \mid \mathcal{E}),$$

where  $\mathcal{E} = \{M > 0, (1/4)Np_N < K < 2Np_N\}$ . With  $\log(N)^2/N \leq p_N \leq N^{-3/5}$ , the first two terms in the right hand side of the above result are  $o(1/N^2)$ , so we must bound the third term, or  $\mathbb{P}(\{D(v)\} = \{D(w)\} \mid \mathcal{E})$ .

For the  $K$  “private” neighbors of  $v$  (neighbors of  $v$  that are not adjacent to  $w$ ), we denote  $v_i$  to be the  $i$ th private neighbor for  $i \in [1, K]$ . Then the degree of  $v_i$  can be written as follows:

$$D_i(v) = V_i + A_i + 1.$$

$V_i$  is the number of edges between  $v_i$  and the set of vertices that are either adjacent to both  $v$  and  $w$  or to neither of them, which has size  $N - 2K - 2$ .  $A_i$  is the number of edges between  $v_i$  and the remaining private vertices  $v_1, \dots, v_k, w_1, \dots, w_k$  excluding  $v_i$  itself. Finally, we add 1 at the end as  $v_i$  is connected to  $v$ .

Similarly, we define

$$D_i(w) = W_i + B_i.$$

We want to show that 1) with sufficiently high probability  $A_i$  and  $B_i$  are bounded by a constant, and 2), the chance that independent binomial multisets are within constants is small. The first condition tells us that with high probability, the range of difference  $A_i - B_i$  is small, which implies that the difference  $D_i(v) - D_i(w)$  is tightly concentrated as well. The second condition tells us the probability that two independent multisets are identical is very small, i.e. the probability of  $\{D(v)\}$  and  $\{D(w)\}$  being the same is very small.

To begin, notice that for given  $M > 0$  and  $K$ , the collections  $\{V_1, \dots, V_K\}$  and  $\{W_1, \dots, W_K\}$  are independent collections of i.i.d  $\text{Bin}(N - 2K - 2, p_N)$  random variables. These random variables are independent of  $A_i, B_i$  for all  $i$ . The  $A_i$ 's and  $B_i$ 's are each marginally distributed  $\text{Bin}(2K - 1, p_N)$ . Thus, using the tail bound from Lemma 2 for binomial distributions:

$$\mathbb{P}(A_i < x, B_i < x, i = 1, \dots, K) \geq 1 - 2K \mathbb{P}(A_1 > x) \geq 1 - 2K \left(\frac{e}{x}\right)^x ((2K - 1)p_N)^x.$$

Setting  $x = 13$  and  $K < 2Np_N$ , we reach our desired result:

$$\mathbb{P}\left(\max_i \{A_i, B_i\} > 13\right) \leq o(N^{-2}).$$

Thus we have shown that with sufficiently high probability,  $A_i$  and  $B_i$  are bounded by a constant. To show the second statement, we work with vectors that count frequency instead of multisets. Define

$$X_i := |\{j : V_j + A_j = i\}|$$

as the number of occurrences where  $V_j + A_j = i$ , and

$$Y_i := |\{j : W_j + B_j = i\}|$$

similarly, for  $i = 0, \dots, N - 2K + 12$ . The original multisets are equal if and only if  $X_i = Y_i$  for all  $i$ . The left side of our desired result,

$$\mathbb{P}(\{V_1 + A_1, \dots, V_K + A_K\} = \{W_1 + B_1, \dots, W_K + B_K\}) = o(1/N^2),$$

can be upper bounded by

$$\mathbb{P}\left(X_{j_i} = Y_{j_i}, i = 0, \dots, \lfloor \alpha \sqrt{Np_N} \rfloor - 1\right)$$

for currently unchosen  $\alpha$ . Note that  $X_{j_i}$  represents the count of indices  $j$  such that  $V_j + A_j = j_i$  for specific  $j_i$ . We have reduced the index range using the fact that most of the values of  $V_j + A_j$  and  $W_j + B_j$  are concentrated in this window. Thus checking this reduced index range provides an upper bound on our desired probability.

For simplicity, define

$$j_i := \lfloor Np_N \rfloor + i,$$

and define the index set

$$\mathcal{I}(N, \alpha) := \left\{0, \dots, \lfloor \alpha \sqrt{Np_N} \rfloor - 1\right\}.$$

We wish to upper bound the probability of  $X_i = Y_i$  for all  $i$  by bounding  $\mathbb{P}(X_{j_i} = Y_{j_i}, i = 0, \dots, \lfloor \alpha \sqrt{Np_N} \rfloor - 1)$ .

To accomplish this, we want to prove that the probability of  $X_{j_i}$  deviating from its expectation is small:

$$\mathbb{P}(X_{j_i} > (1 + \delta)\mathbb{E}X_{j_i}, \text{ for some } i \in \mathcal{I}) = o(N^{-2}).$$

By the local central limit theorem for the binomial distribution, there exist positive constants  $c_1 = c_1(\alpha)$ , where  $c_1$  is a function of  $\alpha$ , and  $c_2$  for all  $i \in \mathcal{I}$  and  $k = 1, \dots, K$ :

$$\frac{c_1}{\sqrt{Np_N}} \leq \mathbb{P}(V_k + A_k = j_i), \quad \mathbb{P}(W_k + B_k = j_i) \leq \frac{c_2}{\sqrt{Np_N}}.$$

Here we put everything together, using that  $X_{j_i}$  is a sum of  $K$  independent Bernoulli variables. Applying the Chernoff bound and a Union bound for independent Bernoulli variables gives:

$$\mathbb{P}(X_{j_i} > (1 + \delta)\mathbb{E}X_{j_i}, \text{ for some } i \in \mathcal{I}) \leq \sum_{i \in \mathcal{I}} 2 \exp \left\{ -\frac{\delta^2}{2 + \delta} \mathbb{E}X_{j_i} \right\}.$$

Lower bounding  $\mathbb{E}X_{j_i}$  from the local central limit theorem result and bounding the sum size gives:

$$\sum_{i \in \mathcal{I}} 2 \exp \left\{ -\frac{\delta^2}{2 + \delta} \mathbb{E}X_{j_i} \right\} \leq 2\alpha \sqrt{Np_N} \exp \left\{ -\frac{\delta^2}{2 + \delta} \cdot \frac{c_1}{3} \sqrt{Np_N} \right\}.$$

Simplifying and recalling the assumption  $p_N \sim N^{-3/5}$  gives:

$$\mathbb{P}(X_{j_i} > (1 + \delta)\mathbb{E}X_{j_i}, \text{ for some } i \in \mathcal{I}) \leq 2\alpha N^{1/5} \cdot N^{-\frac{\delta^2}{2+\delta} \cdot \frac{c_1}{3}} \cdot o(1).$$

To complete the argument, we must bound the probability that two degree multisets, perturbed by small random fluctuations, are identical. We aim to show

$$\mathbb{P}(X_{j_i} = Y_{j_i}, i \in \mathcal{I} \mid X_{j_i} \leq (1 + \delta)\mathbb{E}X_{j_i}, \text{ for all } i \in \mathcal{I}) = o(N^{-2})$$

for a chosen index set  $\mathcal{I}$  and small  $\delta > 0$ .

Let  $K_0 = K$  and  $K_i = K - \sum_{l=0}^{i-1} Y_{j_l}$ , the number of Bernoulli trials remaining after observing  $i$  outcomes in  $\mathcal{I}$ . Then define  $\mathcal{F}_i$  to be the sigma field generated by  $Y_{j_0}, \dots, Y_{j_i}$ , the previous observed counts up to  $i$ . Finally, we note that  $Y_{j_i}$  is the sum of  $K_i$  Bernoulli random variables just like  $X_{j_i}$ , each with success probability  $Q$  bounded as:

$$\frac{c_1}{\sqrt{Np_N}} \leq \frac{c_1/\sqrt{Np_N}}{1 - ic_1/\sqrt{Np_N}} \leq Q \leq \frac{c_2/\sqrt{Np_N}}{1 - ic_2/\sqrt{Np_N}} \leq \frac{c_2}{\sqrt{Np_N}} (1 - \alpha c_2)^{-1}.$$

Thus we can apply the local central limit theorem for the sum of independent Bernoulli random variables, resulting in:

$$\mathbb{P}(Y_{j_i} = X_{j_i} \mid \mathcal{F}_{i-1}) \leq C \left[ K_i \cdot \frac{c_1}{\sqrt{Np_N}} \left( 1 - \frac{c_2}{\sqrt{Np_N}(1 - \alpha c_2)} \right) \right]^{-1/2}$$

for a constant  $C$ . This tells us that the probability that  $Y_{j_i}$  and  $X_{j_i}$  exactly match is at most something like  $\frac{1}{\sqrt{Np_N}}$ . We choose  $\alpha$  small enough and lower bound  $K_i \geq Np_N$  for all  $i$  to conclude that the right hand side above is  $O((Np_N)^{-1/4})$ .

We union bound across all  $i \in \mathcal{I}$ , with  $|\mathcal{I}| \approx \alpha \sqrt{Np_N}$ , and simplify for the assumption that  $p_N \leq N^{-3/5}$ .

$$\begin{aligned} \mathbb{P}(X_{j_i} = Y_{j_i}, i \in \mathcal{I} \mid X_{j_i} \leq (1 + \delta)\mathbb{E}X_{j_i}, \text{ all } i \in \mathcal{I}) &= \exp \left\{ -\frac{\alpha}{4} \sqrt{Np_N} (\log(Np_N) + O(1)) \right\} \\ &\leq \exp \left\{ -\frac{\alpha}{4} \log(N) (\log(\log(N)) + O(1)) \right\} = o(N^{-2}). \end{aligned}$$

Thus we have shown that for vertices  $v, w$  in an Erdős–Rényi graph, the probability that  $\deg(v) = \deg(w)$  is sufficiently small, and the probability that the 2-neighborhoods of  $v$  and  $w$  are equal is sufficiently small as well. Therefore, 3-neighborhoods suffice to identify graphs following the conditions of the theorem with high probability.  $\square$



## 5 Reconstruction

After an in-depth exploration of identifiability, we can zoom back out to the second main question surrounding graph shotgun assembly: reconstruction. We now briefly discuss the threshold results for reconstructibility for different values of  $r$ ; these results are still being improved on by mathematicians.

With reconstructibility, we also take into account the differences between sparse and dense graphs (with small and large  $p$ , respectively), as the probability of reconstructibility from radius  $r$  seems to fluctuate between different phases. Here, however, the threshold occurs when  $p$  is around  $n^{-\frac{2r+1}{2r}}$ . For clarity, we henceforth refer to these different phases dependent on  $p$  as small  $p$  vs. large  $p$  rather than sparse vs. dense. Some results from Johnston, Kronenberg, Roberts, and Scott [4] are as follows:

Let  $G \in G(n, p)$ .

$r = 2$ : If  $p = o(n^{-5/4})$ , then  $G$  is reconstructible from its 2-neighborhoods with high probability.

$r = 3$ : If  $p = o(n^{-7/6})$ , then  $G$  is reconstructible from its 3-neighborhoods with high probability.

Fixed  $r \geq 4$ : If  $p = o(n^{-\frac{2r+1}{2r}})$ , then  $G$  is reconstructible from its  $r$ -neighborhoods with high probability.

Thus if  $p$  is small, growing slower than the  $n^{-\frac{2r+1}{2r}}$  threshold, then  $G$  is reconstructible from its  $r$ -neighborhoods with high probability. However, if  $p$  is larger, growing faster than  $n^{-\frac{2r+1}{2r}}$  but smaller than a second phase transition threshold (specified below for the three cases), then with high probability,  $G$  is not reconstructible.

For  $\alpha > 0$ :

$r = 2$ : If  $p = \omega(n^{-5/4})$  and  $p \leq \frac{1}{3}n^{-3/4}(\log n)^{1/4}$ , then with high probability  $G$  is not reconstructible from its 2-neighborhoods.

$r = 3$ : If  $p = \omega(n^{-7/6})$  and  $p \leq \alpha \frac{\log^2 n}{n(\log \log n)^3}$ , then with high probability  $G$  is not reconstructible from its 3-neighborhoods.

Fixed  $r \geq 4$ : If  $p = \omega(n^{-\frac{2r+1}{2r}})$  and  $p \leq \alpha \frac{\log n}{rn}$ , then with high probability  $G$  is not reconstructible from its  $r$ -neighborhoods.

Once  $p$  is even larger than the second phase transition threshold, reconstructibility becomes likely again. The demonstration of a second phase is quite recent and begs the question of what other new discoveries and progress are yet to be made on this problem.

## 6 Conclusion

The shotgun assembly problem is a fascinating problem in graph theory, with applications ranging from machine learning to medicine. Our main approach to this problem involved two guiding questions of identifiability and reconstruction; we addressed identifiability separately for sparse and dense Erdős-Rényi graphs.

For sparse graphs, the Identifiability Theorem provided us with upper and lower bounds for  $r$  in order to ensure a high probability of identifiability from  $r$ -neighborhoods, and for dense graphs, the 3-neighborhood Theorem provided us with a constant  $r = 3$  which ensures a high probability of identifiability.

On the question of reconstruction, we briefly introduced some recently presented results and how various phases for  $p$  affect reconstructibility. As recently as 2023, new bounds for this problem have been established, and the author is looking forward to seeing more progress on this problem.

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